CHAPTER 5

Series Solutions of ODEs

Power series:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots, \quad |x| < 1$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$$

$$f(x) = 1 + (x-1) + (x-1)^2 + (x-1)^3 + \cdots$$

5.1 Power Series Method

Familiar Power Series

$$\frac{1}{1-x} = \sum_{m=0}^{\infty} x^m = 1 + x + x^2 + \cdots \qquad (|x| < 1, \textbf{ geometric series})$$

$$e^x = \sum_{m=0}^{\infty} \frac{x^m}{m!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\cos x = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{(2m)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \quad + \cdots$$

$$\sin x = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+1}}{(2m+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \quad + \cdots.$$

Q. Solve using power series

$$y' - y = 0$$

Assume: $y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$

where $a_0, a_1, a_2, \ldots$ are unknowns we need to solve

→ strategy: substitute $y = a_0 + a_1 x + \cdots$ in given ODE

→ match like powers of x

$$y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots$$

$$y' - y = 0$$

$$\left[y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots \right]$$

$$y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots$$

$$\left(a_1 + 2a_2 x + 3a_3 x^2 + \cdots \right) - \left(a_0 + a_1 x + a_2 x^2 + \cdots \right) = 0$$

$$(a_1 - a_0) + (2a_2 - a_1) x + (3a_3 - a_2) x^2 + (4a_4 - a_3) x^3 + \cdots = 0$$

by matching coefficients of x^n in LHS & RHS

$$a_1 - a_0 = 0$$

$$2a_2 - a_1 = 0$$

$$3a_3 - a_2 = 0$$

$$4a_4 - a_3 = 0$$

$$a_1 = a_0$$

$$a_2 = \frac{a_1}{2} = \frac{a_0}{2}$$

$$a_3 = \frac{a_2}{3} = \frac{a_0}{3!}$$

$$a_4 = \frac{a_3}{4} = \frac{a_0}{4!}$$

$$\cdots$$

$$a_n = \frac{a_0}{n!}$$

$$y = a_0 + a_0 x + \frac{a_0}{2}x^2 + \frac{a_0}{3!}x^3 + \frac{a_0}{4!}x^4 + \cdots$$

$$= a_0 e^x$$

$$y'' - y' + xy = 0$$

$y = a_0 + a_1 x + a_2 x^2 + \cdots$

→ Substitute power series in given ODE

→ match powers of x in LHS & RHS

→ find $a_0, a_1, a_2, \ldots$ etc

$$y = a_0 + a_1 x + a_2 x^2 + \cdots$$

$$xy = a_0 x + a_1 x^2 + a_2 x^3 + \cdots$$

$$y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots$$

$$y'' = 2a_2 + 6a_3 x + 4.3\, a_4 x^2 + 5.4 a_5 x^3 + \cdots$$

$$y'' - y' + xy = 0$$

$$(2a_2-a_1) + (6a_3-2a_2+a_0)x + (4\cdot 3a_4-3a_3+a_1)x^2 + (5\cdot 4a_5-4a_4+a_2)x^3 + \cdots = 0$$

no initial conditions given:

a_0 & a_1 will be arbitrary

$$2a_2-a_1=0$$

$$a_2=\frac{a_1}{2}$$

$$6a_3-2a_2+a_0=0$$

$$a_3=\frac{2a_2-a_0}{6}=\frac{a_1-a_0}{6}$$

$$12a_4-3a_3+a_1=0$$

$$a_4=\frac{3a_3-a_1}{12}=\frac{\frac{a_1-a_0}{2}-a_1}{12}$$

claim:

$$a_n = \frac{(3-n)a_1 - a_0}{n!}$$

by Abhyudaya

$\forall\ n \geq 2$

check again

solution

find it

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

$$y' = a_1 + 2a_2 x + 3a_3 x^2 + \cdots$$

$$\left.\begin{aligned} y(0) &= 5 \\ y'(0) &= 2 \end{aligned}\right| \begin{aligned} &\rightarrow a_0 = 5 \\ &\quad\;\; a_1 = 2 \end{aligned}$$

Last time:

$$y'' + f(x)y' + g(x)y = q(x)$$

There are theorems which guarantee that solution of ODE exist

Today:

Laplace Transforms